Ex-01:

Problem statement: Write *a* program to (a) display a "?", (b) read two decimal digits whose sum "is less than 10, (c) display them and their sum on the next line, with an appropriate message.

Sample execution:
?27
THE SUM OF *2* AND 7 IS 9

Code

```
.model small
.stack 100h

.data
msg  db 'THE SUM OF $'
msg1 db ' $'
msg2 db 'AND $'
msg3 db 'IS $'

.code
main proc

mov ax,@data
mov ds,ax

; display '?'
mov dl,'?'
mov ah,2
int 21h

; read first digit
mov ah,1
int 21h
mov bl,al

; read second digit
mov ah,1
int 21h
mov cl,al

; new line
mov dl,0dh
```

```
mov ah,2
int 21h
mov dl,0ah
int 21h

; display message
lea dx,msg
mov ah,9
int 21h

; display first digit
mov dl,bl
mov ah,2
int 21h

lea dx,msg2
mov ah,9
int 21h

; display second digit
mov dl,cl
mov ah,2
int 21h

lea dx,msg3
mov ah,9
int 21h

; ---- FIXED SUM PART ----
sub bl,30h   ; ASCII ? number
sub cl,30h

add bl,cl    ; numeric sum

add bl,30h   ; number ? ASCII

mov dl,bl
mov ah,2
int 21h

; exit
mov ah,4ch
int 21h

main endp
end main
```

Output

Ex-02:

Problem statement: Write a program to (a) prompt the user, (b) read first, middle, and last initials of a person's name, and (c) display them down the left margin.

Sample execution:
ENTER THREE INITIALS: JFK
J

F

K

Code

```
.model small
.stack 100h

.data
msg db 'ENTER THREE INITIALS: $'

.code
main proc

mov ax, @data
mov ds, ax
```

```
; display message
lea dx, msg
mov ah, 9
int 21h

; read first character
mov ah, 1
int 21h
mov bl, al

; read second character
mov ah, 1
int 21h
mov cl, al

; read third character
mov ah, 1
int 21h
mov bh, al

; new line
mov ah, 2
mov dl, 0dh
int 21h
mov dl, 0ah
int 21h

; print first initial
mov dl, bl
mov ah, 2
int 21h

; new line
mov dl, 0dh
int 21h
mov dl, 0ah
int 21h

; print second initial
mov dl, cl
mov ah, 2
int 21h

; new line
mov dl, 0dh
int 21h
```

```
mov dl, 0ah
int 21h

; print third initial
mov dl, bh
mov ah, 2
int 21h

; exit
mov ah, 4ch
int 21h

main endp
end main
```

Output

Ex-03:

Problem statement: Write a program to read one of the hex digits *A-F,* and display it on the next line in
decimal. Sample
execution:

ENTER A HEX DIGIT: C
IN DECIMAL *IT* rs 12

Code

```
.model small
.stack 100h

.data
msg1 db 'Enter a hex digit: $'
msg2 db 0dh,0ah,'In decimal it is: $'

.code
main proc

mov ax,@data
mov ds,ax

; display message
lea dx,msg1
mov ah,9
int 21h

; input
mov ah,1
int 21h
mov bl,al

; ---- correct conversion ----
sub bl,37h      ; 'A'→10, 'B'→11, ... 'F'→15

; display result message
lea dx,msg2
mov ah,9
int 21h

; ---- convert to decimal digits ----
mov al,bl
```

```
mov ah,0
mov cl,10
div cl          ; AL = tens, AH = ones

add al,30h      ; ASCII
mov dl,al
mov ah,2
int 21h

mov al,ah
add al,30h
mov dl,al
mov ah,2
int 21h

; exit
mov ah,4ch
int 21h

main endp
end main
```

Output

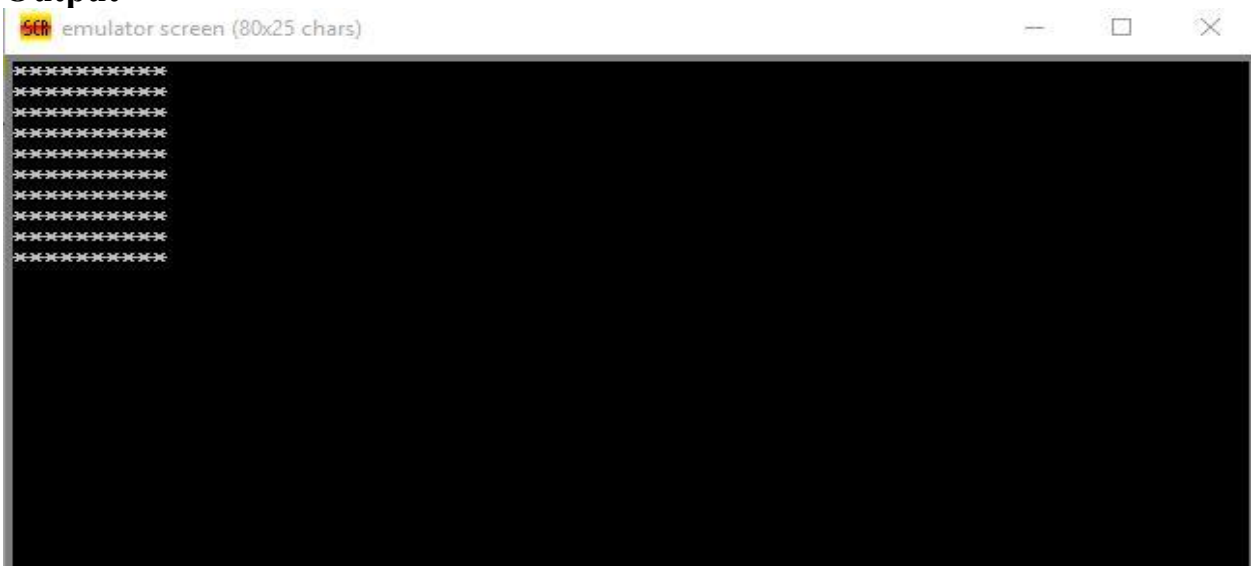
Ex-04:

Problem statement: Write *a* program to display *a* to x 10 solid box of asterisks. *Hint:* declare *a* string in the data segment that specifies the box, and display it with INT 2lh, function 9h.

Code

```
.model small
.stack 100h

.data
msg1 db '**********$'

.code
main proc

mov ax, @data
mov ds, ax

mov cx, 10        ; loop 10 times

print_line:

; print stars
lea dx, msg1
mov ah, 9
int 21h

; new line
mov ah, 2
mov dl, 0dh
int 21h
mov dl, 0ah
int 21h

loop print_line   ; repeat 10 times

; exit
mov ah, 4ch
int 21h

main endp
end main
```

Output



Ex-05:

Problem statement: Write a program to (a) display"?", (b) read three initials,(<;) display them in the middle of an 11 x 11 box of asterix, and (d)
beep the computer.

Code

```
.model small
.stack 100h

.data
top    db '***********$'
mid_l  db '****$'
mid_r  db '****$'

.code
main proc

mov ax,@data
mov ds,ax

; prompt
mov dl,'?'
mov ah,2
```

```
int 21h

; input 3 chars
mov ah,1
int 21h
mov bl,al

mov ah,1
int 21h
mov cl,al

mov ah,1
int 21h
mov bh,al

; newline
mov ah,2
mov dl,0dh
int 21h
mov dl,0ah
int 21h

; ----- top border -----
lea dx,top
mov ah,9
int 21h

; newline
call newline

; ----- upper part -----
mov cx,3
upper:
lea dx,top
mov ah,9
int 21h
call newline
loop upper

; ----- middle line -----
lea dx,mid_l
mov ah,9
int 21h
```

```
mov dl,bl
mov ah,2
int 21h

mov dl,cl
int 21h

mov dl,bh
int 21h

lea dx,mid_r
mov ah,9
int 21h

call newline

; ----- lower part -----
mov cx,4
lower:
lea dx,top
mov ah,9
int 21h
call newline
loop lower

; beep
mov dl,07h
mov ah,2
int 21h

; exit
mov ah,4ch
int 21h

; ----- newline procedure -----
newline proc
mov ah,2
mov dl,0dh
int 21h
mov dl,0ah
int 21h
ret
newline endp
```

```
main endp
end main
```

Output

Ex-06:

Problem statement: Write a program to display a "?", read two capital letters, and display them on the next line In alphabetical order.

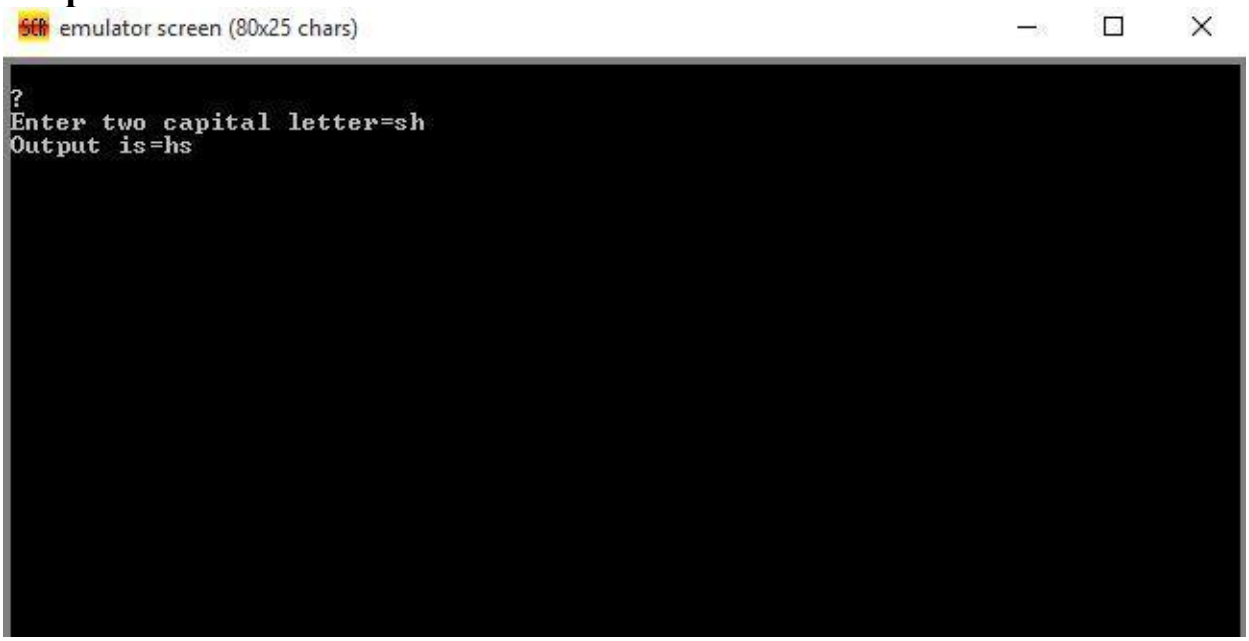
Code

```
.model small
.stack 100h

.data
msg  db 10,13,'?$'
msg1 db 10,13,'Enter two capital letters=$'
msg2 db 10,13,'Output is=$'

.code
main proc

mov ax,@data
```

```
mov ds,ax

; display '?'
lea dx,msg
mov ah,9
int 21h

; prompt
lea dx,msg1
mov ah,9
int 21h

; input first letter
mov ah,1
int 21h
mov bl,al

; input second letter
mov ah,1
int 21h
mov cl,al

; output message
lea dx,msg2
mov ah,9
int 21h

; compare
cmp bl,cl
ja go        ; if BL > CL → swap order

; normal order
mov dl,bl
mov ah,2
int 21h

mov dl,cl
int 21h
jmp exit

go:
mov dl,cl
mov ah,2
int 21h

mov dl,bl
```

```
int 21h

exit:
mov ah,4ch
int 21h

main endp
end main
```

Output



Ex-07:

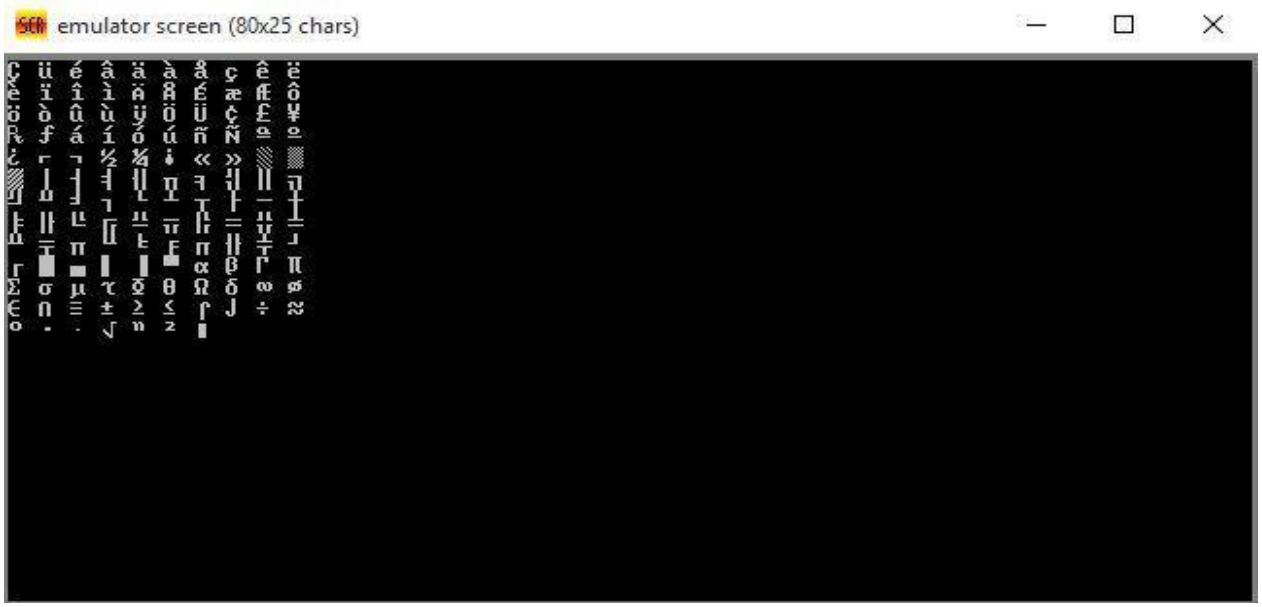
Problem statement: Write *a* program to display the extended ASCII characters (ASCJI codes 80h to FF_h). Display 10 characters per line, separated by blanks. Stop after the extended characters have been displayed once.

Code

```
.model small
.stack 100h
.data

.code
```

```
main proc

mov cx,127     ; starting ASCII
mov bl,0       ; counter for 10 chars

print:

cmp cx,255
ja exit        ; stop after 255

; print character
mov dl,cl
mov ah,2
int 21h

; print space
mov dl,' '
mov ah,2
int 21h

inc cx
inc bl

cmp bl,10
je newline

jmp print

newline:
mov ah,2
mov dl,0dh
int 21h
mov dl,0ah
int 21h

mov bl,0
jmp print

exit:
mov ah,4ch
int 21h

main endp
end main
```

Output



Ex-08:

Problem statement: Write a program that will prompt the user to enter a hex digit character ("0"· ... "9" or "A" ... "F"), display it on the next line in decimal, and ask the user if he or she wants to do it again. If the user types "y" or "Y", the ·program repeats; If the user types anything else, the program terminates. If the user enters an illegal character, prompt the user to try again.

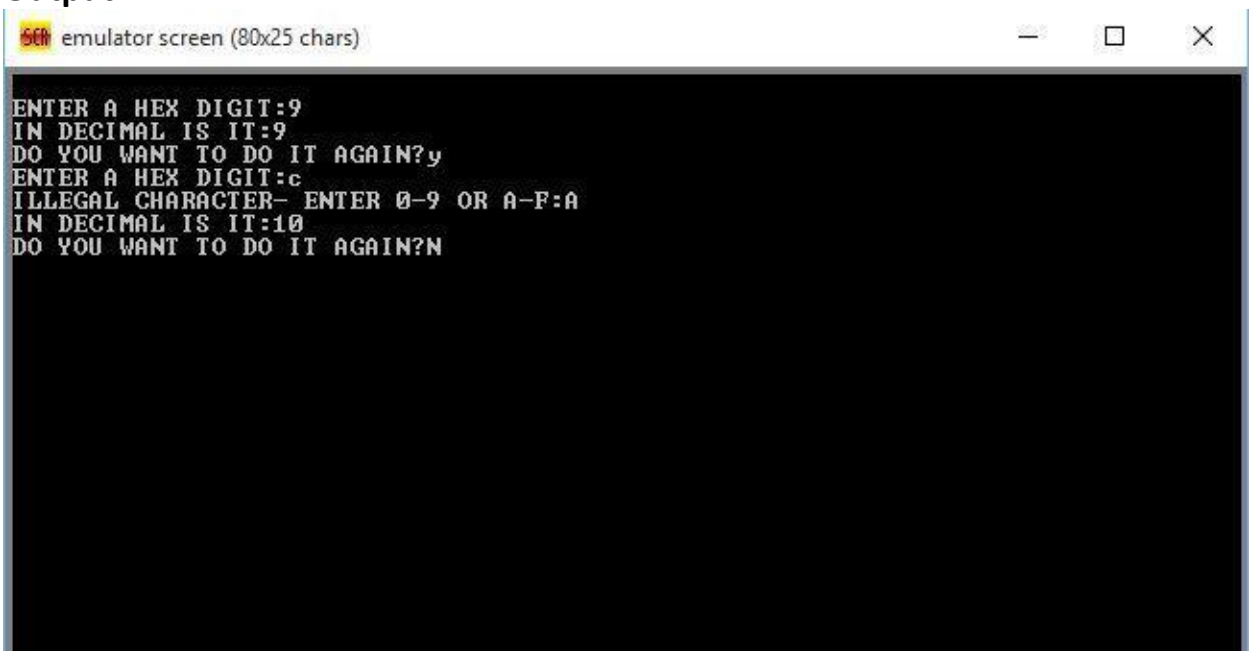
Code

```
.model small
.stack 100h

.data
msg1 db 10,13,'ENTER A HEX DIGIT: $'
msg2 db 10,13,'IN DECIMAL IT IS: $'
msg3 db 10,13,'DO YOU WANT TO DO IT AGAIN? $'
msg4 db 10,13,'ILLEGAL CHARACTER - ENTER 0-9 OR A-F: $'

.code
main proc
```

```
again:
mov ax,@data
mov ds,ax

; prompt
lea dx,msg1
mov ah,9
int 21h

; input
mov ah,1
int 21h
mov bl,al

; ----- check range -----
cmp bl,'0'
jb illegal

cmp bl,'9'
jbe num

cmp bl,'A'
jb illegal

cmp bl,'F'
jbe hex

jmp illegal

; ----- number case -----
num:
lea dx,msg2
mov ah,9
int 21h

mov dl,bl
mov ah,2
int 21h
jmp ask

; ----- hex case -----
hex:
lea dx,msg2
mov ah,9
int 21h

sub bl,37h      ; convert A–F → 10–15
```

```
mov al,bl
mov ah,0
mov cl,10
div cl          ; split digits

add al,30h
mov dl,al
mov ah,2
int 21h

mov al,ah
add al,30h
mov dl,al
mov ah,2
int 21h

jmp ask

; ----- invalid input -----
illegal:
lea dx,msg4
mov ah,9
int 21h
jmp again

; ----- repeat -----
ask:
lea dx,msg3
mov ah,9
int 21h

mov ah,1
int 21h

cmp al,'y'
je again
cmp al,'Y'
je again

exit:
mov ah,4ch
int 21h

main endp
end main
```

Output



Ex-09:

Problem statement: Do programming exercise 10, except that if the user fails to enter a hex-digit character In three tries, display a message and terminate the program.

Code

```
.model small
.stack 100h

.data
msg1 db 10,13,'ENTER A HEX DIGIT: $'
msg2 db 10,13,'IN DECIMAL IT IS: $'
msg3 db 10,13,'DO YOU WANT TO DO IT AGAIN? $'
msg4 db 10,13,'ILLEGAL CHARACTER - ENTER 0-9 OR A-F: $'
msg5 db 10,13,'TOO MANY TRIES $'

.code
main proc

mov ax,@data
mov ds,ax
```

```
mov cx,0        ; retry counter (ONLY once)

again:

; prompt
lea dx,msg1
mov ah,9
int 21h

; input
mov ah,1
int 21h
mov bl,al

; -------- check --------
cmp bl,'0'
jb illegal

cmp bl,'9'
jbe num

cmp bl,'A'
jb illegal

cmp bl,'F'
jbe hex

jmp illegal

; -------- number --------
num:
lea dx,msg2
mov ah,9
int 21h

mov dl,bl
mov ah,2
int 21h
jmp ask

; -------- hex --------
hex:
lea dx,msg2
mov ah,9
int 21h
```

```
sub bl,37h      ; A–F → 10–15

mov al,bl
mov ah,0
mov cl,10
div cl

add al,30h
mov dl,al
mov ah,2
int 21h

mov al,ah
add al,30h
mov dl,al
mov ah,2
int 21h

jmp ask

; -------- invalid --------
illegal:
inc cx
cmp cx,3
je too_many

lea dx,msg4
mov ah,9
int 21h
jmp again

too_many:
lea dx,msg5
mov ah,9
int 21h
jmp exit

; -------- repeat --------
ask:
lea dx,msg3
mov ah,9
int 21h

mov ah,1
int 21h
```

```
cmp al,'y'
je again
cmp al,'Y'
je again

exit:
mov ah,4ch
int 21h

main endp
end main
```

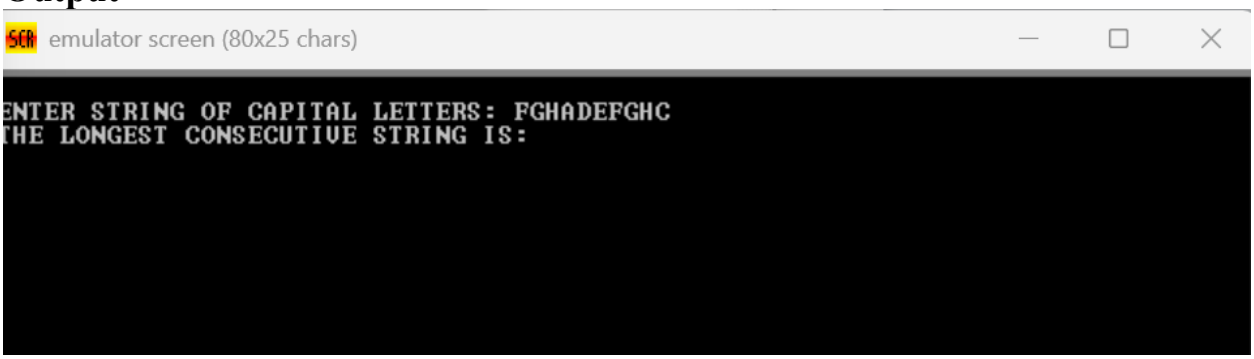
Output

Ex-10:
Problem statement:

Code

```
.model small
.stack 100h

.data
msg1 db 10,13,'ENTER STRING OF CAPITAL LETTERS: $'
msg2 db 10,13,'THE LONGEST CONSECUTIVE STRING IS: $'
buffer db 100 dup(?)

.code
main proc
mov ax,@data
mov ds,ax

; prompt
lea dx,msg1
mov ah,9
int 21h

; -------------------------
; READ STRING
; -------------------------
lea si,buffer
xor cx,cx

read_loop:
mov ah,1
int 21h

cmp al,0Dh
je done_read

mov [si],al
inc si
inc cx
jmp read_loop

done_read:

; -------------------------
; INIT
; -------------------------
```

```
lea si,buffer

xor dx,dx        ; best start index
mov bp,1         ; best length

xor di,di        ; current start index
mov bx,1         ; current length

mov al,[si]
inc si
dec cx
jz print_result

; -------------------------
; SCAN
; -------------------------
scan:
mov ah,[si]

mov dl,al
inc dl
cmp ah,dl
jne reset

inc bx
jmp check

reset:
mov bx,1
mov di,si
sub di,offset buffer

check:
cmp bx,bp
jle skip
mov bp,bx
mov dx,di

skip:
mov al,ah
inc si
loop scan

; -------------------------
; PRINT
; -------------------------
```

```
print_result:

lea dx,msg2
mov ah,9
int 21h

lea si,buffer
add si,dx

mov cx,bp

print_loop:
mov dl,[si]
mov ah,2
int 21h
inc si
loop print_loop

mov ah,4ch
int 21h

main endp
end main
```

Output